

Nonlinear Phase Portraits and Uniqueness Barriers

Thursday, October 29, 2026

Definition: Planar autonomous system

Let $U \subseteq \mathbb{R}^2$ be open, and let $f, g : U \rightarrow \mathbb{R}$. A **planar autonomous system** is

$$\dot{x} = f(x, y), \quad \dot{y} = g(x, y),$$

where $(x, y) \in U$.

Notation: Vector form of a planar system

Write

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \quad \mathbf{x} = (x, y), \quad \mathbf{F} : U \rightarrow \mathbb{R}^2, \quad \mathbf{F}(x, y) = (f(x, y), g(x, y)).$$

Recall: Equilibrium of a planar system

A point $(x^*, y^*) \in U$ is an **equilibrium** if

$$f(x^*, y^*) = g(x^*, y^*) = 0.$$

Definition: Nullclines

The **x -nullcline** is

$$\{(x, y) \in U : f(x, y) = 0\}.$$

The **y -nullcline** is

$$\{(x, y) \in U : g(x, y) = 0\}.$$

Observation: Directions along nullclines

Away from an equilibrium, the vector field is vertical on the x -nullcline and horizontal on the y -nullcline.

Observation: A nullcline need not be invariant

The condition $f(x, y) = 0$ makes the vector field vertical on the x -nullcline, but it does not guarantee that the vector field is tangent to that nullcline. Thus an x -nullcline need not be invariant. The analogous statement holds for a y -nullcline.

Recall: Invariant set

Let Φ be a possibly local flow. A set M is **invariant** if every solution starting in M remains in M throughout its maximal existence interval.

Definition: Positively invariant set

A set M is **positively invariant** if every solution starting at $\mathbf{x}_0 \in M$ satisfies

$$\Phi(t, \mathbf{x}_0) \in M$$

for every $t \geq 0$ in its maximal existence interval.

Question: What can nullclines determine before integration?

Analyze

$$\dot{x} = x + e^{-y}, \quad \dot{y} = -y.$$

Solution. The nullclines are

$$x = -e^{-y}, \quad y = 0.$$

Their unique intersection is the equilibrium $(-1, 0)$. The component signs are

region	sign $\dot{x}$	sign $\dot{y}$
$x > -e^{-y}, y > 0$	+	−
$x < -e^{-y}, y > 0$	−	−
$x > -e^{-y}, y < 0$	+	+
$x < -e^{-y}, y < 0$	−	+

The line $y = 0$ is invariant because $\dot{y} = 0$ there. On that line,

$$\dot{x} = x + 1,$$

so solutions move away from $(-1, 0)$. Also,

$$y(t) = y_0 e^{-t} \rightarrow 0 \quad (t \rightarrow \infty).$$

Thus $(-1, 0)$ repels trajectories along the invariant line $y = 0$. On the other hand, $y(t) \rightarrow 0$ as $t \rightarrow \infty$ for every initial value, so trajectories are attracted toward the line $y = 0$ in the y -coordinate.

Visualization: Nullcline sign map for the worked nonlinear system

	$x < -e^{-y}$	$x > -e^{-y}$
$y > 0$		
$y < 0$		

On $x = -e^{-y}$,

$$y > 0 : \downarrow, \quad y < 0 : \uparrow.$$

On $y = 0$,

$$x < -1 : \leftarrow, \quad x > -1 : \rightarrow.$$

Away from $x = -e^{-y}$, trajectory slopes satisfy

$$\frac{dy}{dx} = \frac{-y}{x + e^{-y}}.$$

Definition: Direction field

A **direction field** records the direction of $\mathbf{F}(\mathbf{x})$ at each state where $\mathbf{F}(\mathbf{x}) \neq \mathbf{0}$, but suppresses its magnitude.

Recall: Classical fourth-order Runge–Kutta method

Let $\mathbf{x}_n$ approximate a solution of $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$ at time t_n , and let $h > 0$. The **classical fourth-order Runge–Kutta method** defines

$$\mathbf{k}_1 = \mathbf{F}(\mathbf{x}_n),$$

$$\mathbf{k}_2 = \mathbf{F}(\mathbf{x}_n + \frac{h}{2}\mathbf{k}_1),$$

$$\mathbf{k}_3 = \mathbf{F}(\mathbf{x}_n + \frac{h}{2}\mathbf{k}_2),$$

$$\mathbf{k}_4 = \mathbf{F}(\mathbf{x}_n + h\mathbf{k}_3),$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4).$$

Theorem: Local existence and uniqueness

Let $D \subseteq \mathbb{R}^n$ be open and let $\mathbf{F} : D \rightarrow \mathbb{R}^n$ be continuously differentiable. For every $\mathbf{x}_0 \in D$, the initial-value problem

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \quad \mathbf{x}(0) = \mathbf{x}_0,$$

has a unique solution on some open interval containing 0.

Corollary: Distinct maximal trajectories do not cross

Under the hypotheses of the local existence-and-uniqueness theorem, let $\mathbf{x}_j : I_j \rightarrow D$, $j = 1, 2$, be maximal solutions, meaning that neither solution extends to a larger time interval. Their trajectory images are either disjoint or identical.

Proof. Suppose

$$\mathbf{x}_1(t_1) = \mathbf{x}_2(t_2) = \mathbf{p}.$$

Define

$$\mathbf{y}_1(s) = \mathbf{x}_1(t_1 + s), \quad \mathbf{y}_2(s) = \mathbf{x}_2(t_2 + s).$$

Autonomy implies that both solve the same system with initial condition $\mathbf{y}_j(0) = \mathbf{p}$. Uniqueness gives $\mathbf{y}_1 = \mathbf{y}_2$ on the intersection of their shifted time intervals. If one shifted interval extended beyond the other, the longer solution would extend the shorter one, contradicting maximality. Hence the shifted intervals and the solution curves coincide, so the original solutions have the same trajectory image with different time origins.

Proposition: Regular invariant-curve test

Let $D \subseteq \mathbb{R}^2$ be open, let $\mathbf{F} : D \rightarrow \mathbb{R}^2$ be locally Lipschitz, and let

$$C = \{(x, y) \in D : h(x, y) = 0\},$$

where $h \in C^1(D)$ and $\nabla h \neq \mathbf{0}$ on C . Suppose

$$\nabla h(x, y) \cdot \mathbf{F}(x, y) = 0 \quad \text{for every } (x, y) \in C.$$

Then C is invariant under $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$.

Proof. The regular-level-set theorem makes C locally a smooth curve. The displayed condition says that $\mathbf{F}$ is tangent to C . Restricting the vector field to C produces a solution that remains in C . Uniqueness forces the full-system solution with the same initial point to coincide with that curve-following solution.

Question: Can a trajectory cross an invariant circle?

Consider

$$\dot{x} = -y + x(1 - x^2 - y^2), \quad \dot{y} = x + y(1 - x^2 - y^2).$$

Find the polar equations and classify the unit circle.

Solution. Let $r^2 = x^2 + y^2$. Then

$$\begin{aligned} \frac{d}{dt}r^2 &= 2x\dot{x} + 2y\dot{y} \\ &= 2(x^2 + y^2)(1 - x^2 - y^2) \\ &= 2r^2(1 - r^2). \end{aligned}$$

For $r > 0$,

$$\dot{r} = r(1 - r^2).$$

Question: Can a trajectory cross an invariant circle? (continued)

Also,

$$\dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} = 1.$$

Hence

$$0 < r < 1 \Rightarrow \dot{r} > 0, \quad r = 1 \Rightarrow \dot{r} = 0, \quad r > 1 \Rightarrow \dot{r} < 0.$$

The unit circle is an invariant periodic orbit of period 2π . Every nonzero trajectory inside it spirals outward toward it, and every trajectory outside it spirals inward toward it. Uniqueness prevents any distinct trajectory from crossing it.

Proposition: A closed orbit is a barrier

Let C be a simple closed trajectory of a planar system with unique solutions. A trajectory beginning in the bounded component of $\mathbb{R}^2 \setminus C$ cannot enter the unbounded component while it exists.

Proof. By the Jordan curve theorem, every continuous path from the bounded component to the unbounded component meets C . Such an intersection would make the new trajectory coincide with C by uniqueness, which is impossible for a trajectory that began off C .

Question: Does local existence imply existence for all future time?

Solve

$$\dot{x} = x^2, \quad x(0) = x_0 > 0.$$

Solution. Separation gives

$$-\frac{1}{x(t)} = t + C.$$

The initial condition yields $C = -1/x_0$, so

$$x(t) = \frac{x_0}{1 - x_0 t}.$$

The right-hand side of the ODE is smooth on $\mathbb{R}$, but the solution becomes unbounded as $t \uparrow 1/x_0$. Thus local uniqueness does not imply global existence.

Theorem: Continuation from compact containment

Let $\mathbf{F} : D \rightarrow \mathbb{R}^n$ be locally Lipschitz on an open set D , and let $\mathbf{x} : [0, T_{\max}) \rightarrow D$ be a maximal solution. If $\mathbf{x}([0, T_{\max}))$ lies in a compact subset of D , then $T_{\max} = \infty$.

Equivalently, if $T_{\max} < \infty$, then for every compact set $K \subseteq D$ there exists $t_K < T_{\max}$ such that

$$\mathbf{x}(t) \notin K \quad \text{whenever } t_K < t < T_{\max}.$$

Idea: Analytical and numerical phase portraits

Analytically justified phase-portrait information often comes from

$$\boxed{\text{equilibria}} + \boxed{\text{nullclines and signs}} + \boxed{\text{invariant sets}} + \boxed{\text{uniqueness barriers}}.$$

Observation: A numerical trajectory is not a proof

A numerical trajectory is an approximation depending on its initial state, time interval, method, and step size. It can reveal a candidate invariant boundary or closed orbit, but it does not prove invariance, nonintersection, periodicity, or an asymptotic limit.